

Layer-Wise Analysis of Emotion and Speaker Information in WavLM Speech Representations

1 Project Title

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2 Student Information

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3 Project Overview

This project will investigate which layers of a frozen WavLM Base+ speech model contain the most useful information for categorical emotional-expression recognition, and how strongly that information is associated with speaker identity. Using the speech-only portion of RAVDESS, the study will classify happy, sad, angry, and neutral acted expressions. A compact traditional baseline based on MFCC, pitch, and energy features will provide an interpretable comparison. Emotion recognition will use actor-independent evaluation, while a separate speaker-identification probe will measure how accessible actor identity is in each representation. WavLM will be used as a frozen feature extractor; it will not be trained or fine-tuned.

The main research question is: Which layers of frozen WavLM contain the most useful information for categorical emotional-expression recognition, and how strongly is that information entangled with speaker identity? The secondary question is: How does the best WavLM layer compare with traditional MFCC, pitch, and energy features for emotion recognition and speaker identification?

4 Background and Motivation

Speech conveys linguistic content as well as acoustic cues associated with emotional expression. Pitch, energy, timing, and spectral characteristics can change across expressive categories, but these cues also vary naturally between speakers. As a result, a system may exploit actor-specific speaking patterns instead of learning features that generalise to unseen speakers.

Traditional speech emotion recognition (SER) studies commonly compare hand-crafted acoustic features and classifiers (El Ayadi et al., 2011). Recent systems also use self-supervised learning (SSL) speech models, which provide learned representations from raw waveforms. Broad comparisons reported in recent research, including Pratihast (2025), identify WavLM and UniSpeech-SAT as promising representations for SER. This project therefore focuses on one model, WavLM Base+, and studies how emotion and speaker information vary across its layers instead of conducting a large model or classifier comparison.

The project addresses an active question in SER: whether representations that make emotion easy to classify also retain strong speaker identity information. Speaker information is not assumed to be

entirely harmful, because a speaker's habitual pitch and intensity range may help interpret expressive changes. Measuring both types of information can show how they co-occur across layers. Recent speaker-aware SER work explicitly models speaker and emotion information and studies their disentanglement (Shi et al., 2025). The EMO-SUPERB benchmark provides further context for controlled, reproducible evaluation of frozen speech representations (Wu et al., 2024).

RAVDESS contains acted vocal expressions. Accordingly, this project will describe its target as categorical emotional-expression recognition and will not interpret the labels as direct measurements of a speaker's internal emotional state. The work connects classical signal processing—framing, windowing, STFT, Mel analysis, MFCC, pitch, and energy—to modern learned speech representations.

5 Proposed Methodology

Tools and platforms

MATLAB and the ELEC5305 Toolbox will be used for traditional audio analysis and feature extraction. Python with PyTorch and Hugging Face Transformers will load the pretrained Microsoft WavLM Base+ model and extract hidden representations. MATLAB or Python with scikit-learn will be used to train simple linear classifiers and calculate evaluation metrics. The model will remain frozen, with no fine-tuning.

Data source and labels

The only dataset will be the public Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) (Livingstone & Russo, 2018). Speech-only recordings will be used; song recordings will be excluded. Four acted emotional-expression categories will be selected: happy, sad, angry, and neutral. Emotion and actor labels will be parsed from the audio filenames. The study will use the 24 actors represented in the dataset and will document the data preparation and label mapping.

Traditional acoustic baseline

Each recording will be converted to mono and resampled to 16 kHz where required. Audio will be divided into overlapping frames using a Hamming window. The classical processing path will include the short-time Fourier transform (STFT), Mel spectrum, and MFCC extraction. The compact fixed-length feature vector will include MFCC statistics, F0 statistics (including median and range), RMS or short-time energy, duration, and optionally spectral centroid. Mean and standard deviation across frames will be used where appropriate. This focused feature set will retain an interpretable signal-processing baseline without expanding into a large collection of handcrafted descriptors.

Frozen WavLM representations

The pretrained microsoft/wavlm-base-plus model will receive 16-kHz speech waveforms. Hidden representations will be extracted from selected transformer layers, provisionally layers 1, 4, 8, and 12; if computational resources allow, all layers will be examined. Each layer produces a sequence over time. Mean pooling across the time dimension will produce one fixed-dimensional embedding per utterance. Embeddings will be cached to make the probing experiments reproducible and efficient.

Emotion and speaker probes

The same simple classifier family—linear SVM or logistic regression—will be applied to the traditional

feature vector and to each WavLM layer embedding. Feature scaling and model selection will be conducted using training data only. The primary emotion metric will be macro-F1, with accuracy, per-class precision and recall, and confusion matrices reported as supporting results.

- Emotion probe: predict happy, sad, angry, or neutral using actor-independent folds. Actors in an emotion test fold will not appear in its training data. Several actor-level folds will be used where feasible.
- Speaker probe: predict actor identity from each representation using a separate classifier. For this diagnostic task, utterances from each actor will be divided between probe training and test sets, with no utterance shared across the split. Speaker-probe results will be reported separately from emotion-recognition results.

For every representation, emotion performance and speaker-identification accuracy will be compared. The analysis will examine whether layers with stronger emotion performance also make actor identity more accessible. No assumption will be made in advance that speaker information must reduce emotion performance.

Acoustic interpretation and figures

To connect model behaviour to physical signal properties, distributions of median F0, F0 range, RMS energy, duration, and (if retained) spectral centroid will be compared across the four expression categories. Representative waveforms, spectrograms, or pitch contours may be used to illustrate relevant differences. Planned figures include: emotion macro-F1 by WavLM layer with the traditional baseline; speaker-identification accuracy by layer; an emotion-versus-speaker performance plot labelled by layer; acoustic-feature distributions; and confusion matrices for the traditional baseline and best WavLM layer.

Evaluation and reproducibility

All folds, preprocessing choices, random seeds, and classifier settings will be documented. Emotion and speaker probe splits will follow their respective protocols described above. The repository will include RAVDESS access instructions, label mapping, split definitions, feature extraction, WavLM embedding extraction, evaluation scripts, cached embeddings where storage permits, final figures, and instructions to reproduce the analysis. RAVDESS audio will not be redistributed unless its terms permit it.

Optional extension

If the core experiments are complete, early, middle, and late UniSpeech-SAT layers may be probed using the same protocol to check whether the observed pattern is specific to WavLM. This extension is not required for project completion. A CNN extension and additional datasets are outside the planned scope.

6 Expected Outcomes

The project is expected to deliver a reproducible analysis of emotion and speaker information in traditional acoustic features and frozen WavLM representations. The results will establish which selected WavLM layers make acted emotional expressions most accessible to a simple classifier and how speaker-identification performance changes across those layers. The comparison will describe the relationship between emotion performance and speaker information without assuming that one

must decrease as the other increases.

The main deliverables will include:

- A MATLAB-based traditional feature pipeline for MFCC, F0, energy, duration, and optional spectral centroid.
- A reproducible frozen WavLM Base+ hidden-state extraction and mean-pooling pipeline.
- Actor-independent emotion-probe results for the four RAVDESS speech-expression classes.
- Separate speaker-identification probe results for each traditional or WavLM representation.
- Layer-wise plots of emotion macro-F1 and speaker-identification accuracy, plus an emotion-versus-speaker comparison.
- Acoustic feature analyses and confusion matrices comparing the traditional baseline with the best WavLM layer.
- Documented code, configuration, dataset instructions, and reproduction steps in the GitHub repository.

7 Timeline (Weeks 1–13)

Weeks	Planned work
1–2	Finalise the research question, review key SER and SSL literature, and update the repository plan.
3–4	Download and organise RAVDESS speech recordings; implement the four-class mapping and actor-level emotion folds.
5–6	Implement the compact MFCC, F0, energy, and duration baseline; inspect acoustic distributions.
7–8	Load frozen WavLM Base+ and extract/cache selected layer embeddings; verify preprocessing and pooling.
9–10	Run actor-independent emotion probes and separate speaker-identification probes for all representations.
11	Compare layer-wise results; produce emotion, speaker, and entanglement plots.
12	Interpret acoustic patterns, prepare baseline-versus-best-layer confusion matrices, and discuss limitations.
13	Complete the report, repository documentation, and reproducibility checks; undertake the optional extension only if time allows.

8 References

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